SatQuery AI

Satellite Image Query System — Completed Work Summary

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SatQuery AI answers natural-language questions over real satellite imagery through a six-stage pipeline: ingestion and sensor normalisation, an agentic controller with a trained query router, six specialist model heads, calibrated answer integration with honest abstention, an evidence and trace PDF layer, and a FastAPI backend driving a live 3-D globe frontend. This report inventories everything completed and verified to date, with the measured numbers from the project's own evaluation artifacts.

STATUS AT A GLANCE

6 / 6	6 / 3	10	4
pipeline stages complete	model intents trained vs required	real satellite sites ingested	public VQA benchmarks

All six pipeline stages are implemented and demonstrably working on real pixels. Four public remote-sensing vision-language VQA benchmarks (RSVQA-LR, RSVQA-HR, VRSBench, CDVQA) were evaluated end to end in both zero-shot and supervised settings, alongside ablation, calibration, change-mask and optical+SAR fusion studies. The backbone is fine-tuned twice over — EuroSAT (97.7% replay accuracy during joint training) and BigEarthNet.txt S2 optical + S1 SAR — and the query router is on its second trained generation.

DELIVERED COMPONENTS

Component	Status	Evidence / measured result
Stage 0 — ingest, sensor profiles, QA mask, tile cache	**Complete**	10 named sites, bi-temporal S2 + S1 where covered (6 of 10 carry S1; 4 remain optical-only pending SAR backfill); pair alignment 4.18 px → 0.14 px after repair
Stage 1 — agentic controller (router v2, gate, tool registry)	**Complete**	Router v2: 84.9% 5-fold CV, 100% on the 13-query eval suite, all six intents routed correctly
Stage 2 — backbone + six specialist heads (vqa, describe, ground, change, change-qa, fuse)	**Complete**	All six intents produce real output on real Assam pixels; fuse executes at 77.5% confidence with gate 5/5
Stage 3 — calibration, conflict detection, abstention	**Complete**	Scorecard: pipeline 100% on answered queries, honest abstention 15.4% (e.g. water questions on a dry site abstain by design)
Stage 4 — trace + evidence + PDF report	**Complete**	Per-run TraceRecord rendered to PDF (demo artifact at docs/Prototype/demo_report.pdf)

Component	Status	Evidence / measured result
Stage 5 — FastAPI backend + live frontend wiring	**Complete**	POST /query answers in under 1 s (cached); globe frontend auto-probes /health and switches to live mode
3-D globe frontend (docs/index.html)	**Complete + fixed**	Half-ingested-imagery blackout fixed via 12-sector completeness gate (threshold 0.8) and region probe with date step-back; zoom view verified 0% black
Evaluation harness	**Complete**	run_eval.py scorecard + eval_benchmarks.py on three public benchmarks (see companion benchmark report)

Table 1 — Component inventory with measured evidence. Status reflects the latest verification pass (latest verification pass).

TRAINING COMPLETED

Backbone fine-tune 1 — EuroSAT (Sentinel-2)

The ResNet-50 self-supervised Sentinel-2 backbone was fine-tuned on EuroSAT (13-band, 10 classes, 16,200 / 5,400 / 5,400 train-val-test split) on the project's single 6 GB GPU. That initial fine-tune reached **97.78%** held-out test accuracy; the later joint passes then re-tuned the encoder, so this figure is historical — the current encoder's EuroSAT linear-probe is ~94.9%, and 97.7% survives as the replay-accuracy check during joint training (below). The checkpoint (models/backbone_ft.pth, 202 MB) is auto-loaded by get_backbone().

Backbone fine-tune 2 — BigEarthNet.txt, S2 + S1

The prescribed BigEarthNet.txt corpus was used without the 170+ GB LMDB: the 0.47 GB annotation parquet was paired with torchgeo multipart tars and 7,084 patches selectively streamed (round-robin by country/snow), of which 6,861 exist on disk after source-side quality filtering. The optical encoder was fine-tuned on BEN-19 multi-label BCE, then feature-cached heads were trained on 87,452 VQA examples (validation 62.2%). The same patches' VV/VH SAR complements were extracted through the s1_name mapping (0 errors) and a SAR encoder was fine-tuned with the optical path frozen (validation loss 0.254), then saved back into the same checkpoint (657 keys).

Joint fine-tune 3 — encoder-unfrozen multi-task pass (v2)

A joint multi-task run (scripts/train_joint.py) then unfroze both encoders and trained one loop over four tasks: BigEarthNet.txt VQA through the full encoder (not cached features), BEN-19 land-cover on S2 and on S1, and a EuroSAT replay that keeps core land-cover competence (97.7% replay accuracy; the re-tuned encoder's own EuroSAT linear-probe is ~94.9%). Text moved from TF-IDF to MiniLM-L6 sentence embeddings (any phrasing embeds meaningfully) with a FiLM option-scoring head trained jointly with the encoder. The corpus was expanded to 17,084 patches (16,000 train / 1,084 bench) and trained in two passes; band-dropout and RGB-stretch input modes teach the encoder the partial-band and stretched-RGB inputs the live pipeline and public benchmarks present. A third pass (September 2026) continued training on the BigEarthNet.txt S2+S1 tasks alone and lifted held-out validation VQA accuracy to **76.41%** (from 62.2% for the frozen-encoder v1), with every supervised benchmark at RSVQA-LR 80.83%, VRSBench 83.84%, CDVQA 77.18%. The scorer now answers free-form live queries — it is wired into the /query pipeline as satquery-joint-v2 and reconciled against

the spectral measurement in every answer.

Router v2 — six-intent query classifier

The TF-IDF + logistic-regression router was retrained on a corpus grown from 59 to 393 queries (counting, single-direction change and comparison phrasings included in the latest revision), balanced across six intents with eval queries excluded. Five-fold cross-validation accuracy is 84.4%, and it routes every query in the 13-query evaluation suite correctly, including the previously misrouted present-tense land-cover question.

Adapted benchmark scorers

Supervised scorers for each public benchmark were trained and persisted (models saved under data/bench/): rsvqa_lr_adapted.pkl, vrsbench_adapted.pkl and cdvqa_adapted.pkl. With the joint v2 stack (see below) the supervised results are RSVQA-LR 80.83%, VRSBench 83.84% and CDVQA **77.18%** — the CDVQA scorer uses change-vector features (difference and sum of the two scene embeddings) with a pair-aware train/validation split, which raised its test accuracy from a chance-level 55.5% — details in the companion benchmark report.

EXTENDED EVALUATION AND VALIDATION (SEPTEMBER 2026)

Beyond the headline VQA benchmarks, a second evaluation wave measured the system's harder design claims directly — four ablations, confidence calibration, change-mask localisation, per-class optical+SAR fusion, out-of-distribution captioning and grounding, and a high-resolution VQA set. Each writes a machine-readable JSON that drives the companion benchmark report (Tables 4–10); the honest, sometimes-bounded outcomes follow.

Study	Measured result	Verdict
Optical+SAR fusion under cloud (ablation 1)	at 80% cloud, fusion holds 64% water recall vs 17% optical-only	**supported**
Trained router v2 vs rule router v1 (ablation 3)	v2 89.5% vs v1 71.0% on 38 held-out paraphrases (+18.4 pts)	**supported**
Confidence calibration (ECE)	adapted-scorer confidence is informative (AUROC 0.81) and temperature scaling cuts ECE 0.066→0.021; zero-shot softmax confidence is near-uninformative (AUROC ≈ 0.5)	**supported**
Optical+SAR per-class IoU, BigEarthNet held-out	max-fusion macro-mIoU 0.322 vs 0.295 best single sensor at the deployed 0.5 threshold; tuned optical-only recovers most of it	**bounded**
Change-mask localisation, LEVIR-CD (RGB)	unsupervised CVA F1 0.085 / IoU 0.044 on binary building-change — a lower bound; the real edge is multispectral, not 3-band CVA	**bounded**
VRSBench captioning & grounding (coverage)	BLEU-4 0.000, CIDEr-D 0.002 — the land-cover CaptionHead is out-of-distribution on scene prose; grounding vocabulary does not overlap (reported, not hidden)	**bounded (OOD)**

Study	Measured result	Verdict
RSVQA-HR high-resolution aerial VQA	zero-shot 41.0% / supervised 85.6% on 222,684 test questions	**supported**

Table 4 — Extended evaluation summary. Full tables, per-class breakdowns and methodology are in the companion benchmark report (docs/SatQuery_AI_Benchmark_Results.pdf, Tables 4–10). Results are reported honestly, including where an outcome is bounded rather than a clean win.

FIXES FROM VERIFICATION PASSES

Issue	Fix
Globe back hemisphere rendered flat / washed-out	Root cause was a half-ingested GIBS product day (VIIRS granules arrive by longitude band over ~20 h; WMS still returns HTTP 200 all-black JPEGs). The composite layer now tracks per-longitude completeness in 12 sectors and only accepts a day whose worst sector is at least 80% covered, stepping back a day otherwise, with a max-worst fallback. Front hemisphere brightness mean rose from 106 to 179 and the header now names the stepped-back date.
Zoom view went black while zooming	The pass/zoom fetcher now probes the requested bounding box at low resolution before the full fetch and steps back to the most recent date whose frame over that region has content. The HUD shows the actual captured date. Verified 0% black pixels during a zoom pass.
Cloud-shadow QA mask flagged dark water and mangrove as shadow	Shadow now additionally requires blue-scattering brightness against the local neighbourhood. Sundarbans detects 38.2% water, matching the NDWI ground truth exactly.
Fuse query executed as describe	The pipeline had passed the stale pre-repair pair-check to the compatibility gate. Re-running the check after alignment repair fixed it; fuse now executes as fuse at 77.5% confidence.
CDVQA supervised accuracy stuck near chance	Random row splits leaked identical image-pair features across folds, and raw concatenated features made the difference inaccessible to a small MLP. Group-aware splits keyed by pair id plus change-vector features fixed it (see benchmark report).

Table 2 — Issues found during live verification and their resolutions.

DATA AND ARTIFACTS ON DISK

Artifact	Contents
data/raw/ + manifest	Real satellite imagery for 10 sites (assam, delhi, sundarbans, punjab, gangotri, rondonia, creekfire, nile, dubai, kaziranga), bi-temporal S2 + S1 where covered
models/backbone_ft.pth	Fine-tuned optical + SAR encoders and heads (657 keys)

Artifact	Contents
models/router_v2.pkl	Trained six-intent router with metrics dict
data/bench/	CDVQA / RSVQA-LR / VRSBench shards, feature caches and adapted scorers used for the benchmark report
data/*.json (bench, ablation, calibration, ben_fuse_iou, levir_cd, rsvqa_hr, vrsbench caption/grounding, scorecard)	Machine-readable evaluation results backing every number in this report and the companion benchmark report
docs/index.html + START_DEMO.bat	Live 3-D globe frontend with backend auto-discovery, and the one-click demo launcher

Table 3 — Principal artifacts. Evaluation inputs were streamed selectively; both source tars were deleted after extraction (~260 GB free afterwards).

HOW TO RUN THE DEMO

Step	Command / action
Start backend	`python -m uvicorn satquery.api:app --port 7860`
Open demo	`Double-click START_DEMO.bat (opens docs/index.html and starts the backend)`
Ask a query	`Type into the query box; the frontend switches to live mode when /health responds`
Re-run evaluation	`python scripts/run_eval.py --quick`
Reproduce benchmarks	`python scripts/eval_benchmarks.py --bench all --mode both`

The globe, zoom pass and workspace views fetch live NASA GIBS imagery, so an internet connection is required for the Earth view; the analyst pipeline runs fully offline once imagery is cached.